Adaptive four-block contraction

A sharper asymptotic certificate beyond the current SubexponentialSAT research packets

Research prepared for Vladimir Reshetnikov
with ChatGPT

October 26, 2026 (Pacific time)

Abstract

We review twenty refinement packets and the two primary manuscripts. The five newest packets arrived through `docs/incoming` at snapshot `fa6e904d109b`; the closing snapshot `dde01c4a63b8` files them under SubexponentialSAT. The requested improvement in the logarithmic order of general SAT is not established. We do obtain two new explicit exponential upper-bound certificates in the common normalized tensor-contraction model.

For s Boolean wires, maximum degree at most $\lceil s^{9/10} \rceil$, and the specified entry-oracle and precision interface, our cached algorithm has

$$T \le \mathrm{poly}(s+\mathcal{L}) 2^{(C_T+o(1))s/\sqrt{\log_2 s}}, \qquad S \le \mathrm{poly}(s+\mathcal{L}) 2^{(C_S+o(1))s/\sqrt{\log_2 s}},$$

where

$$C_T = \frac{\pi}{\ln 2}\left(\frac{\sqrt{105}}{2} + \sqrt{\frac{183}{40}}\right) = 32.9158138649\ldots, \qquad C_S = \frac{\pi}{\ln 2}\sqrt{\frac{244}{5}} = 31.6617067862\ldots.$$

The best displayed incoming cached certificate is 35.3545880584 in time and 34.2748030050 in space. A second construction preserves polynomial space and has time coefficient 50.3937257005, compared with the best displayed 55.7094783189 and a stronger matched composition 54.1282952066.

The new ingredient is a sharp four-block inequality for adaptive matrix contraction, together with an adaptive four-cluster separator inequality. The choices depend on exact mode-list sizes. Both are proved by elementary finite-dimensional arguments; no compact-mode summation hypothesis is assumed. The global schedules pay for every child contraction and preserve the inherited packing and precision conditions. Supporting code verifies the new finite identities and bounds over exact arithmetic. It does not implement the full SAT algorithm or the asymptotically fast matrix backend. The SAT orders $n \log\log n/\sqrt{\log n}$ and $n/\sqrt{\log n}$ remain unchanged. All SAT consequences retain the explicitly listed upstream deformation and reduction dependencies.

Contents

1 Results and the comparison being made **3**
1.1 What the requested order improvement would require 3

2 Frozen review of the recent commits **4**
2.1 Trust boundary of the source collection 4

3 Normalized input and inherited exact interfaces **5**

4 Exact mode lists and a Euclidean entropy bound **5**
4.1 Counting, indexing, and making adaptive choices 6

5 Exact matrix multiplication and table storage **7**

6 An elementary certificate for adaptive four-block contraction **7**

7 Building the four tables and paying for child calls **9**

8 A complete cached schedule **10**
8.1 Packing, depth, and integer floors 11
8.2 Precision and entry evaluation 11
8.3 Unrolling the recurrence without losing child costs 12

9 A new adaptive separator bound in polynomial space **12**
9.1 The four-cluster inequality 12
9.2 An admissible schedule and its full coefficient 13

10 Comparison with compatible prior constructions **14**
10.1 A slightly faster matrix backend 14

11 Why the stronger logarithmic targets remain unresolved **15**
11.1 Precision peeling is locally useful but not a new leading saving 15
11.2 A short index is not a bounded-degree tensor description 15
11.3 The exterior-power route needs more than reduced degree 15

12 Validation and reproduction **16**

A The twenty-packet review inventory **17**

1 Results and the comparison being made

The useful outcome is a proved change in how the mode graph is contracted. The algorithm chooses among three ways of pairing four blocks after the exact mode budgets have been fixed. Treating this choice as part of the algorithm, rather than bounding a fixed contraction tree by its worst allocation, reduces the leading coefficient.

Theorem 1.1 (Cached normalized contraction certificate)**.** *Assume the normalized interface in Section 3 and the imported identities listed there. There is a deterministic exact contraction algorithm with time coefficient C_T and space coefficient C_S displayed in the abstract. In particular, eventual coefficients* 33 *for time and* 32 *for space are valid. The algorithm uses subexponential space.*

Theorem 1.2 (Polynomial-space certificate)**.** *Under the same interface, there is a deterministic exact contraction algorithm using polynomial space and time*

$$\operatorname{poly}(s+\mathcal{L})2^{(C_P+o(1))s/\sqrt{\log_2 s}}, \qquad C_P = 50.3937257005\ldots.$$

An exact expression for C_P is given in (8); in particular, the eventual coefficient 51 *is valid.*

These are refinements of upper bounds. They are not measured ratios of running times. The common size parameter s counts normalized Boolean network wires. It is not the SAT variable count n. An input reduction and its wire constants are needed to obtain numerical coefficients in terms of n.

Certificate at degree $\lceil s^{9/10} \rceil$	Time	Space
Best displayed incoming cached bound [8]	35.354588	34.274803
Prior compatible cached composition, Section 10	34.832153	33.992701
New cached bound, Theorem 1.1	32.915814	31.661707
Best displayed incoming polynomial-space bound [7]	55.709478	polynomial
Prior ingredients on our polynomial-space schedule	54.128295	polynomial
New polynomial-space bound, Theorem 1.2	50.393726	polynomial

The comparison deliberately includes compatible older ingredients. We do not claim a global optimum over every retuning of every earlier algorithm. The new sharp local inequality is strictly stronger than the fixed-tree and adaptive-crossing estimates used in the supplied packets. The report provides both historical comparisons and matched analytical ablations.

1.1 What the requested order improvement would require

The strongest justified general-density envelope in the reviewed reports is

$$\operatorname{poly}_k(n,m)\, 2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1}$$

It reduces to $2^{O_k(n\log\log n/\sqrt{\log n})}$ in the worst case and $2^{O_{k,D}(n/\sqrt{\log n})}$ at fixed density $m \le Dn$. The new engines retain these orders under the existing reductions.

The conditional $n\log\log n/\log n$ target in the precision packet [4] would be a stronger order improvement. Its structured summation algorithm is not supplied by that packet or by this report. We do not count an unproved conditional proposal as an established upper bound, and we do not claim to outperform it if its missing hypothesis is eventually proved. Section 11 records the unsuccessful order routes investigated here and their precise obstacles.

2 Frozen review of the recent commits

The closing main-branch snapshot is `dde01c4a63b822c25b8eb345299346c36f837b49`, committed on October 26, 2026 at 19:42:25 Pacific time. The opening review used `fa6e904d109b`, committed at 19:31:45, and inspected all five incoming archives. The initially cloned default branch had an additional merge, `fd0edd17e6a2`, with identical relevant source trees. A closing refresh found the placement commit below. The reviewed files were compared with their filed versions, and the provenance ledger retains both locations. No repository change was submitted.

Commit	Relevant change
`2b925d8800cd`	Created the SubexponentialSAT project and filed the seven first-round packets and the primary manuscripts.
`3e096089911f`	Vendored the unchanged OpenAI preprint from `openai/math@7ef5c0cc5c`.
`45233238aebf`	Summarized the strongest first-round claims and separated conditional proposals from established refinements.
`3e57705cf14e`, `d098ba436b33`	Received the eight second-round archives.
`6d3adc450615`	Filed all eight under the research directory; the project README records that their SAT order is unchanged.
`f42479bca245`, `f898a9fde4c3`, `fa6e904d109b`	Added the five further incoming archives reviewed here. These must be included in a current comparison; the opening filed README alone omitted them.
`450252e9d86c`	Filed those five packets unchanged, updated the project summary, and restored two round-two support scripts. The closing `docs/incoming` contains only its README.

The closing check verified all 230 filed members against the five reviewed archives and their Git blobs. There were no omissions or extra members. The two restored scripts match their earlier manifest hashes and concern PDF builds and archive checks, not mathematical algorithms. The provenance ledger records Git blob IDs, SHA-256 hashes, immutable links, and the members inspected inside the five ZIPs. The review inventory is in `provenance/corpus_review.json`. Appendix A gives the mathematical summary. The review covers the main claims and interfaces, not a new formal verification of every upstream proof.

2.1 Trust boundary of the source collection

The project README explicitly says that these are external research packets filed as delivered; it reports no Lean or Rocq development for this project inside ProveIt. The OpenAI manuscript and the separate Claude manuscript are primary claims, not a proof supplied by this report. The upstream Lean scope described in the repository has an additional logarithmic factor compared with the manuscript's Appendix D. We do not equate those two statements or claim to have rebuilt that formalization.

The new theorems below are ordinary mathematical refinements relative to the exact-contraction framework. Their genuinely new combinatorial lemmas are self-contained. The implementation checks are exact finite checks of those components, not substitutes for the uniform asymptotic proofs.

3 Normalized input and inherited exact interfaces

Let $s \geq 2$, $\lambda = \lceil \log_2 s \rceil$, and let $\mathcal{L}$ be the total graph and tensor-description length. The input is a bipartite Boolean tensor network with at most s wires and maximum degree at most $\Delta = \left\lceil s^{9/10} \right\rceil$. Its contraction is

$$Z = \sum_{x \in \{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

The entries are in $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$, with $P_0, \ell \leq s$. Fixed multiples of s give the same leading constants. An original entry routine has time $\mathrm{poly}(s + \mathcal{L})2^{O(s^{9/10})}$ and polynomial space, with fixed interface constants. Arbitrarily expensive tensor oracles are excluded.

During contraction the coefficient ring may be extended to

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell), \qquad \deg f = O(\log s),$$

where f is monic and irreducible modulo two. Each ring entry has polynomially many coordinates of polynomial bit length. The creation precision of every tensor is part of its immutable description. A later call evaluates that tensor at its creation precision and uses the same canonical coordinate representatives. Canonical lifting is not treated as a ring homomorphism.

The following exact ingredients are imported from the preprint and the listed refinements [3, 5, 6, 7, 8].

1. The polynomial Cauchy deformation, its integral evaluation at the chosen nodes, exact Hermite recovery, and the child-network identity.
2. Asymmetric packet packing: with packet length $L = 2^a$, $Q = r^a$ labels per side, row and column degree bounds R, C, nearly lossless padding holds under $QR\,\mathrm{polylog}(s) = o(M)$, $Q^2C\,\mathrm{polylog}(s) = o(M)$, and $LQ^2 = o(M)$. At the root the stronger symmetric term $Q^2\Delta\,\mathrm{polylog}(s) = o(M)$ is sufficient. Padded mass is $(1 + o(1))M$, and each child has $(2/r + o(1))M$ wires.
3. Below the root, the old side retains degree at most Δ; the newly created side has degree at most $2^{a_{d-1}+1}$.
4. Full crossing-block Cauchy modes, each local to one directed pair of groups, with weights

   $$w_q = a + 1 - i + 4q - 2s_2(q), \qquad q \geq 0.$$

   At valuation level i, put $D = 2^{a-i}$. Each directed pair has at most $(2/r^2 + o(1))M/D$ modes at any given index q.
5. Sharp recovery precision $p_t \leq (6 + o(1))MD/u$ once the inherited precision is made negligible by an additive logarithmic buffer.

We prove that our schedules satisfy these hypotheses. We also prove the new mode-graph contraction identities and their resource bounds, rather than assuming a faster oracle for the group functions.

4 Exact mode lists and a Euclidean entropy bound

Fix one active call, interpolation node, and exact array of directed-pair budgets b_{gh} with $\sum_{g \neq h} b_{gh} < p_t$. The number of arrays is polynomial in s because r is a fixed constant. Let $\mathcal{I}_{gh}$ be the set of mode-exponent tuples on that pair of exact weight b_{gh}. An empty list makes this array contribute zero.

Otherwise write

$$h_{gh} = \log_2 |\mathcal{I}_{gh}| \geq 0, \qquad K(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

The contribution is exactly

$$\sum_{(\xi_{gh})\in\prod_{g\neq h}\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{2}$$

Here an entry of Z_g is obtained by contracting the prescribed child network. It is a paid recursive call.

Lemma 4.1 (Uniform entropy ball)**.** *For fixed r, uniformly over the exact arrays and active calls,*

$$\|h\|_2 \leq \left(\frac{1}{r} + o(1)\right) K(M,u).$$

Consequently, for any nonnegative vector z, including one chosen from the actual lists, $z \cdot h \leq (\|z\|_2/r + o(1))K$ when $\|z\|_2 = O(1)$.

Proof. Lower the actual weights to $\widehat{w}_q = 1 + 4q - 2s_2(q)$ and extend the index range. Its counting function is $x/4 + O(\log(x+2))$, giving

$$\sum_{q\geq 0} -\ln(1 - e^{-\theta\widehat{w}_q}) = \frac{\pi^2}{24\theta} + O(\log^2(2 + 1/\theta))$$

as $\theta \downarrow 0$, with a corresponding uniform upper bound for all $\theta > 0$. The identity $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$ supplies the coefficient.

For any $z \in [0,1]^{r(r-1)}$, use parameter θ/z_e for pair e, omitting a pair with $z_e = 0$, and multiply its logarithmic bound by z_e. With $c = (2/r^2 + o(1))M/D$, the result is

$$\ln 2\,(z \cdot h) \leq \theta p_t + \frac{c\pi^2}{24\theta}\|z\|_2^2 + O_r(c\log^2(2 + 1/\theta)).$$

Take $\|z\|_2 = 1$ and optimize the two leading terms. Using $p_t \leq (6 + o(1))MD/u$ yields $(K/r + o(K))$ after division by $\ln 2$. The saddle has order $\sqrt{u}/D$. Our schedules give $D \geq \lambda^6$ up to a fixed factor, $u \leq \lambda$, and $\log D = O(\lambda)$, so the error relative to $M/\sqrt{u}$ is $O(\lambda^{5/2}/D) = o(1)$. This estimate is uniform for every such z, even near a coordinate face. Set $z = h/\|h\|_2$ if $h \neq 0$. The zero case is immediate. □

There is also a way to apply the later arguments using only the source's fixed-vector entropy lemma. The four-block proof produces just 216 possible mean coefficient vectors. Apply the inherited estimate separately to those fixed vectors and take the largest of their finitely many error terms. The separator construction likewise produces a finite family of fixed masks for each fixed arity. Thus the adaptive use does not require an unproved extension of the source estimate to arbitrary variable weights.

4.1 Counting, indexing, and making adaptive choices

For mode weights $w_1, \ldots, w_v$, the dynamic program

$$A(j,b) = A(j-1,b) + A(j,b-w_j), \qquad A(0,0) = 1$$

with zero values at negative budgets counts the exact lists. It supports rank, unrank, and enumeration with polynomial delay. Both the number of modes and the budgets are polynomial in s; counts are bounded by $(p_t+1)^v$ and therefore have polynomial bit length. No counting oracle is needed.

The six aggregated domain sizes for four blocks are exact products of integer list sizes. With matrix exponent 12/5, the fifth power of a merge cost is an integer quotient of products of those sizes. It can be compared by cross multiplication. Alternatively, use ceiling binary logarithms, computed by integer bit length: this changes the cost by only $O_r(1)$, which is negligible in every displayed leading exponent. Thus adaptation needs neither approximate real comparisons nor an unconstructed optimization algorithm.

5 Exact matrix multiplication and table storage

The main theorem uses the same safe exponent $\omega = 12/5$ as the incoming cached-tree packet [8]. The primary matrix-multiplication literature provides an exponent strictly below 12/5 [11]. We need the exact arithmetic bridge, since R_p has nonunits and nilpotents.

Fix a finite rational bilinear multiplication algorithm whose recursive exponent is strictly below 12/5. Exclude the fixed finite set of primes dividing its coefficients' denominators. A product of integer coordinate matrices with shared dimension N and entries below 2^p has entries below $N2^{2p}$. Compute the product over polynomially many nonexceptional small prime fields until their product exceeds this bound, reconstruct by CRT, and reduce modulo 2^p. The required number of primes and all intermediate bit lengths are polynomial in s, because p and $\log N$ are polynomial in s. Expanding ring entries in the fixed module basis requires only polynomially many coordinate products; the integer structure constants reconstruct their ring product. No division by a nonunit in R_p occurs.

A rectangular product with dimensions N_1, N_2, N_3 can be split into square blocks of side comparable to $\min N_j$. The bit work is bounded by

$$\operatorname{poly}(s + \mathcal{L}) N_1 N_2 N_3 (\min N_j)^{-3/5}.$$

For index logarithms x, y, z, its exponent is therefore

$$x + y + z - \tfrac{3}{5} \min(x, y, z). \tag{3}$$

A fixed bilinear base algorithm can process its recursive products sequentially and accumulate outputs, with quadratic field-element storage. The rectangular and CRT wrappers require a polynomial factor times the input/output array sizes. Those sizes are bounded by the work exponent in (3).

Table records have polynomial bit length. The required orderings can be obtained by sorting and scanning records in polynomial overhead per record. The argument does not assume free random access to integers with exponentially many bits. The finite code uses ordinary exact multiplication to test the tensor identity, not the theoretical fast backend.

6 An elementary certificate for adaptive four-block contraction

Let the six edges of the complete graph on four vertices have nonnegative weights w_e. These weights are logarithms of index-domain sizes; opposite directions have already been aggregated. Suppose a rectangular matrix product with index logarithms x, y, z costs, up to polynomial factors,

$$2^{x+y+z-(3/5)\min\{x,y,z\}}.$$

For each of the three perfect matchings $M = \{ab, cd\}$, put $C_M = w_{ac} + w_{ad} + w_{bc} + w_{bd}$. Contract the pair ab and the pair cd independently, then take the inner product of their boundary tables. The two

merge exponents are

$$C_M + w_{ab} - \tfrac{3}{5}\min\{w_{ab}, w_{ac} + w_{ad}, w_{bc} + w_{bd}\},$$
$$C_M + w_{cd} - \tfrac{3}{5}\min\{w_{cd}, w_{ac} + w_{bc}, w_{ad} + w_{bd}\}.$$

Their maximum dominates the final inner-product exponent C_M. Let $F_M(w)$ be this maximum and $F(w) = \min_M F_M(w)$.

Theorem 6.1 (Sharp four-block Euclidean certificate)**.** *For every nonnegative $w \in \mathbb{R}^6$,*

$$F(w) \le \sqrt{\frac{244}{75}}\,\|w\|_2.$$

The numerical constant is best possible for these three contraction choices and this rectangular matrix-product cost model.

Proof. Each F_M is the maximum of six linear forms. For either internal edge $e \in M$, one form has coefficient vector

$$A_e = \mathbf{1}_{E\setminus M} + \tfrac{2}{5}\mathbf{1}_e.$$

The other two forms are obtained by replacing it by

$$B_e = A_e + \tfrac{3}{5}(\mathbf{1}_e - \mathbf{1}_f - \mathbf{1}_g),$$

where f, g are the two crossing edges incident to one endpoint of e. All coefficient vectors are nonnegative.

Choose one of the six forms from each matching, arbitrarily. Temporarily replace every selected B form by its corresponding A form. Since every edge is crossing for exactly two matchings and the three chosen internal edges are distinct, the sum of the resulting vectors is

$$b = 2\mathbf{1} + \tfrac{2}{5}z, \qquad z \in \{0,1\}^6, \quad \sum_e z_e = 3.$$

Consequently $\|b\|_2^2 = 3(12/5)^2 + 3 \cdot 2^2 = 732/25$.

Suppose $k \le 3$ selected forms were of type B, and let $D = \sum(\mathbf{1}_e - \mathbf{1}_f - \mathbf{1}_g)$ be the sum of their correction vectors. We have $\sum_e D_e = -k$. Let $N \ge 0$ count, with multiplicity, occurrences of a negative correction on one of the three edges selected by z. Every positive correction occurs on such an edge, so $z \cdot D = k - N$. It follows that

$$b \cdot D = -\tfrac{8}{5}k - \tfrac{2}{5}N \le -\tfrac{8}{5}k.$$

Each individual correction has squared norm three. For corrections from different matchings, their positive edges are distinct and their negative two-edge sets intersect in at most one edge. Their inner product is therefore at most one. Hence

$$\|D\|_2^2 \le 3k + 2\binom{k}{2} = k(k+2) \le 5k.$$

Restoring the B forms gives a sum vector $b + (3/5)D$, whose norm obeys

$$\begin{aligned}\|b + \tfrac{3}{5}D\|_2^2 - \|b\|_2^2 &= \tfrac{6}{5}b \cdot D + \tfrac{9}{25}\|D\|_2^2 \\ &\le -\tfrac{48}{25}k + \tfrac{45}{25}k \le 0.\end{aligned}$$

Thus the mean of any one-form-per-matching selection has squared norm at most $(732/25)/9 = 244/75$.

For the given w, choose a maximizing form from each matching. The minimum of the resulting three values is at most their mean. Cauchy–Schwarz and the preceding uniform estimate prove $F(w) \le \sqrt{244/75}\,\|w\|_2$.

For sharpness, give the three edges incident to one vertex weight $4/5$ and the other three edges weight $2/3$. Every matching contains one edge of each weight. Its crossing sum is $44/15$, and its larger merge cost is $44/15 + (2/5)(4/5) = 244/75$. The weight vector has squared norm $244/75$, so equality holds. □

For an application with $r = 4b$ original groups divided into four blocks of size b, let $h_{gh} \ge 0$ denote directed-pair index logarithms. Each undirected block-pair weight sums $2b^2$ such logarithms, and the six sets are disjoint. Therefore

$$\|w\|_2^2 \le 2b^2 \sum_{g \ne h} h_{gh}^2.$$

If the established weighted entropy estimate implies $\|h\|_2 \le (1 + o(1))K/r$, the final four-block contraction has coefficient

$$\sqrt{\frac{244}{75}}\,\frac{\sqrt{2}b}{r} = \sqrt{\frac{61}{150}}$$

on the scale K. Formation of the four input block tables, paid recursive oracle evaluations, storage, packing conditions, and the complete recursion remain separate obligations. The theorem by itself makes no SAT complexity claim.

Dependence on the matrix-multiplication exponent. The same argument works for every $7/3 \le \omega \le 3$ when the rectangular cost is $x + y + z - (3 - \omega)\min\{x, y, z\}$. Put $\nu = \omega - 2$. Then $A_e = \mathbf{1}_{E \setminus M} + \nu \mathbf{1}_e$, and a type-$B$ correction is $(1 - \nu)(\mathbf{1}_e - \mathbf{1}_f - \mathbf{1}_g)$. The base sum is $b = 2\mathbf{1} + \nu z$, giving

$$\|b\|_2^2 = 3(2 + \nu)^2 + 12 = 3\omega^2 + 12, \qquad b \cdot D \le -(2 - \nu)k.$$

As above $\|D\|_2^2 \le 5k$, so restoring the corrections changes the squared norm by at most

$$\left[-2(1 - \nu)(2 - \nu) + 5(1 - \nu)^2\right]k = (1 - \nu)(1 - 3\nu)k \le 0.$$

Thus the sharp local constant is $\sqrt{(\omega^2 + 4)/3}$, attained by three star edges of weight $\omega/3$ and three other edges of weight $2/3$. The equal-four-cluster entropy coefficient becomes

$$\sqrt{\frac{\omega^2 + 4}{24}}.$$

This is a parametric contraction theorem; use of a particular ω still requires an established exact arithmetic implementation of the corresponding matrix multiplication bound.

7 Building the four tables and paying for child calls

For $r \in \{4, 8, 12\}$, first cache every Z_g on its incident pair indices. There are $2(r - 1)$ incident directed pairs, so Lemma 4.1 bounds the child-call count by

$$2^{(\alpha_r + o(1))K}, \qquad \alpha_r = \frac{\sqrt{2(r - 1)}}{r}.$$

The fixed factor r and all exact weight arrays are polynomial overhead.

Partition the groups into four equal, predetermined blocks. For any subset B, its table T_B is indexed by its boundary directed pairs and sums the product of the cached factors over the internal pairs. If disjoint sets of sizes a, b are merged in a graph on r groups, the three index classes of the matrix product have cardinalities

$$2a(r-a-b), \qquad 2ab, \qquad 2b(r-a-b).$$

If these numbers are v_1, v_2, v_3, the weighted entropy estimate and (3) give coefficient

$$\frac{1}{r}\sqrt{v_1+v_2+v_3-\frac{21}{25}\min(v_1,v_2,v_3)}. \tag{4}$$

Indeed the cost is the maximum of three linear forms with coefficients $1, 1, 2/5$ on the index classes; apply the entropy ball to each.

For $r = 8$, each initial block has size two. Its classes are $(12, 2, 12)$, and its squared numerator is $608/25$. For $r = 12$, first form pairs and then triples. The size-three merge has classes $(18, 4, 36)$ and squared numerator $1366/25$; smaller merges cost less. For $r = 4$, the four input blocks are the already cached singleton factors. Thus

$$\frac{\sqrt{608/25}}{8} < \beta, \qquad \frac{\sqrt{1366/25}}{12} < \beta, \qquad \beta = \sqrt{\frac{61}{150}}.$$

The new adaptive four-block theorem controls the remaining contraction by $2^{(\beta+o(1))K}$ work. It also bounds the chosen pair tables and matrix temporaries, since every input and output size is at most that matching's merge-cost bound. The fixed initial block tables fit within the same envelope.

Proposition 7.1 (Paid local recurrence)**.** *Suppressing polynomial factors per call, the new cached engine satisfies*

$$T_d \le 2^{(\alpha_{r_d}+o(1))K_d} T_{d+1} + 2^{(\beta+o(1))K_d}, \qquad S_{d,\mathrm{local}} \le 2^{(\beta+o(1))K_d}. \tag{5}$$

All recursive group entries are charged by the first term. The second term uses cached group values and makes no uncharged child call.

The identities hold in any commutative ring: the regrouping changes only the order of finite sums and products. Fast matrix multiplication is an optional implementation of those same products. Taking minima over the three pairings does not alter the value of the contraction.

8 A complete cached schedule

For sufficiently large s, put

$$\Lambda = \lambda - \left\lceil \sqrt{\lambda} \right\rceil, \qquad b_* = \lceil 6 \log_2 \lambda \rceil, \qquad H = \lceil s/\lambda \rceil.$$

Handle the finite smaller range by direct summation. Set $u_d = \lfloor f_d \Lambda \rfloor$ and $a_d = \sum_{j=0}^{d} (u_j + b_*)$. Use

d	r_d	f_d	$A_d = \sum_{j \le d} f_j$	m_d
0	8	$1/60$	$1/60$	1
1	12	$1/90$	$1/36$	$1/4$
2	8	$1/180$	$1/30$	$1/24$
$3+k$	4	$(1/160)(5/8)^k$	$< 1/20$	$(1/96)2^{-k}$

(6)

Here m_d is the limiting wire-count multiplier; the bounds will show $M_d \le (1+o(1))^d m_d s$ uniformly on active paths.

8.1 Packing, depth, and integer floors

At the root, $2A_0 \log_2 r_0 + 9/10 = 1$. The retreat from λ to Λ gives a negative $\Theta(\sqrt{\lambda})$ in its binary-log packing exponent. Below the root, the dominant old-side exponents are

$$\frac{9}{10} + \frac{\log_2 12}{36} < 1 \quad (d = 1), \qquad \frac{9}{10} + \frac{3}{30} = 1 \quad (d = 2), \qquad \frac{9}{10} + 2A_d \le 1 \quad (d \ge 3).$$

Equality at the limiting power is again accompanied by the negative $\Theta(\sqrt{\lambda})$ retreat. For a rational strict certificate at depth one, $12^{100} < 2^{359}$ implies $\log_2 12 < 359/100$, leaving a margin of $1/3600$.

The fresh-side term has leading power $2A_d \log_2 r_d + A_{d-1}$, and rounding has leading power $(1 + 2\log_2 r_d)A_d$. Both are at most $1/4$ throughout this schedule (with a smaller bound at depth one). Additive buffers, polynomial logarithmic factors, and $s/M \le \lambda$ cost only $O((\log \lambda)^2)$ in their binary logarithms. They are absorbed by the strict power gaps or by the $\sqrt{\lambda}$ retreat. New packet degrees are at most $s^{1/20+o(1)}$, below the maintained $s^{9/10}$ cap.

For a noncircular depth argument, provisionally consider $O(\log \lambda)$ depths with tail length at most $\log_{20/11} \lambda + O(1)$. At all those depths the smallest taper is

$$\lambda^{\eta+o(1)}, \qquad \eta = 1 - \frac{\ln(8/5)}{\ln(20/11)} > 0,$$

since $5/8 > 11/20$. Hence all u_d diverge uniformly and the floors are relative $o(1)$ perturbations. The packing inequalities just proved are uniform on this provisional range. Their child ratios are eventually at most $11/20$, with sharper limits $1/4, 1/6, 1/4, 1/2$. Therefore every active path fits inside the provisional range. This closes the bootstrap.

Padding errors are exponentially small in $\sqrt{\lambda}$ or a fixed negative power of s, after polynomial logarithmic factors. Their product over $O(\log \lambda)$ levels is $1 + o(1)$. The asserted m_d bounds and uniform geometric estimates follow.

8.2 Precision and entry evaluation

The inherited precision is $O(s\lambda^2 2^{a_{d-1}})$. For retained levels $i < u_d$, the additive buffer gives $D = 2^{a_d - i} \ge 2^{a_{d-1}+b_*+1}$. Thus

$$\frac{p_{\text{inherited}}}{MD/u_d} = O(\lambda^4 2^{-b_*}) = o(1).$$

The remaining jet-loss term is also lower order. The sharp budget $p_t \le (6 + o(1))MD/u_d$ is therefore uniform. All precisions, interpolation degrees, extension ranks, and tensor-description sizes are polynomial in s.

Leaves cost $2^{O(s/\lambda)}$. Original entry routines cost $2^{O(s^{9/10})} \operatorname{poly}(s + \mathcal{L})$. Packet routines cost at most $\exp(O(2^{a_d} \log s)) = \exp(s^{1/20+o(1)})$ times polynomial factors. Each is $2^{o(s/\sqrt{\lambda})}$. Creation-time representatives are retained; the routine for an old tensor is not replaced by a recursively expanding computation at each precision increase.

8.3 Unrolling the recurrence without losing child costs

Put $\kappa = \pi\sqrt{2}/\ln 2$ and $\gamma = \sqrt{2/5}$. On the scale $s/\sqrt{\lambda}$, the four leading K values are

$$\begin{aligned} k_0 &= \kappa\sqrt{60} = 49.6495177683\ldots, \\ k_1 &= \kappa\sqrt{90}/4 = 15.2019980635\ldots, \\ k_2 &= \kappa\sqrt{180}/24 = 3.5831453061\ldots, \\ k_{3+j} &= (\kappa\sqrt{160}/96)\gamma^j = 0.8445554480\ldots\gamma^j. \end{aligned}$$

Unrolling (5) takes the maximum of each local aggregation cost plus the preceding cache costs. The four finite starting candidates and the infinite cache limit are

Candidate	Coefficient
βk_0	31.6617067862
$\alpha_8 k_0 + \beta k_1$	32.9158138649
$\alpha_8 k_0 + \alpha_{12} k_1 + \beta k_2$	31.4483967553
$\alpha_8 k_0 + \alpha_{12} k_1 + \alpha_8 k_2 + \beta k_3$	31.3778492206
$\alpha_8 k_0 + \alpha_{12} k_1 + \alpha_8 k_2 + \alpha_4 k_3/(1-\gamma)$	32.2464016431

The tail candidates have the form

$$P + \frac{\alpha_4 k_3}{1-\gamma} + \left(\beta - \frac{\alpha_4}{1-\gamma}\right) k_3 \gamma^j,$$

so their maximum is an endpoint or their limit. The second displayed candidate is the largest; exact rational interval certificates in the code verify every strict comparison. It equals the stated C_T.

Local stored tables have coefficient at most βk_d. Ancestor storage adds, rather than multiplies: child calls are processed one at a time while the partially filled parent cache is retained. Since k_d is largest at the root, the space coefficient is $C_S = \beta k_0$. Polynomial work accumulated over $O(\log \lambda)$ depths contributes $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the time exponent. This proves Theorem 1.1.

Here the per-level multiplicative overhead is polynomial in s, not a fresh power of the entire original description length. Immutable tensor descriptions reference original entry routines, and packet routines have size polynomial in s. A single fixed polynomial in $s + \mathcal{L}$ pays for original input handling and an original entry evaluation. Factoring it out of the recurrence avoids a spurious $\mathcal{L}^{O(\log \lambda)}$ factor. This is the same description convention as the imported interface.

9 A new adaptive separator bound in polynomial space

Caching is not needed for a second, independent improvement. Condition on the directed indices crossing a split, contract its two sides separately, multiply their values, and sum over the crossing assignment. This is exact distributivity over finite sums. A depth-first implementation stores only the current assignments, ring accumulators, and recursion stack.

9.1 The four-cluster inequality

For a four-vertex graph let $w_{ij} \geq 0$ be the combined weight of both directions and let $W = \sum w_{ij}$. For a perfect matching $\{e, f\}$, the largest root-to-leaf conditioning cost is $W - \min(w_e, w_f)$. For the three

opposite-edge pairs write a_i for the smaller weight and b_i for the larger. Choosing the matching with largest a_i gives

$$\begin{aligned} W - \max_i a_i &\le \tfrac{2}{3}\sum_i a_i + \sum_i b_i \\ &\le \sqrt{\tfrac{13}{3}}\left(\sum_e w_e^2\right)^{1/2}. \end{aligned}$$

The constant is sharp: set every $a_i = 2$ and every $b_i = 3$.

For four predetermined clusters each of size b, each combined cluster pair contains $2b^2$ directed coordinates. Cauchy–Schwarz bounds the squared norm of their six aggregated weights by $2b^2$ times the directed cross-cluster squared norm. Internal coordinates are disjoint. If $g(r)$ bounds the squared norm of a complete root-to-leaf path functional on r groups, the adaptive construction satisfies

$$g(4b) \le \frac{26}{3}b^2 + g(b). \tag{7}$$

With $g(1) = 0$, this gives $g(4^j) = (26/45)((4^j)^2 - 1)$. In particular

$$g(4) = 26/3, \qquad g(8) = 110/3, \qquad g(16) = 442/3.$$

For $r = 8$, use four clusters of size two and $g(2) = 2$. Fixed balanced binary trees have the larger respective values $10, 42, 170$.

The cost bound for the full adaptive tree follows by combining its cross-cluster and within-cluster norm bounds on disjoint coordinates. For the actual entropy vector, the local time and number of child calls are at most

$$2^{(\sigma_r + o(1))K}, \qquad \sigma_r = \sqrt{g(r)}/r.$$

There are only constantly many cluster choices for fixed r. Select the matching by exact products of list sizes. Every leaf factor evaluation is an actual recursive child call, and is charged by the same path bound.

9.2 An admissible schedule and its full coefficient

Use the same Λ, b_*, H conventions, now with

d	r_d	f_d	A_d	m_d
0	16	$1/80$	$1/80$	1
1	16	$1/80$	$1/40$	$1/8$
2	8	$1/120$	$1/30$	$1/64$
$3+k$	4	$(1/160)(5/8)^k$	$< 1/20$	$(1/256)2^{-k}$.

The root symmetric gate and the subsequent old-side gates have limiting exponent one, with the negative $\sqrt{\lambda}$ retreat. Fresh-side and rounding exponents are at most $1/4$. All arities are powers of two; there is no irrational packing-capacity comparison. The previous depth, precision, floor, and entry-cost argument applies verbatim.

The streamed recurrence is multiplicative, $T_d \le \mathrm{poly}(s)2^{(\sigma_{r_d} + o(1))K_d}T_{d+1}$, so its coefficients sum. The resulting exact expression is

$$C_P = \frac{\pi\sqrt{2}}{\ln 2}\left[\frac{9}{128}\sqrt{80 \cdot \frac{442}{3}} + \frac{1}{512}\sqrt{120 \cdot \frac{110}{3}} + \frac{\sqrt{160 \cdot (26/3)}}{1024(1 - \sqrt{2/5})}\right]. \tag{8}$$

This is $50.3937257005\ldots$. Replace $442/3, 110/3, 26/3$ by $170, 42, 10$ in the same expression to obtain the matched prior coefficient $54.1282952066\ldots$. This separates the gain from the new adaptive inequality from any gain due merely to schedule selection.

All mode indices, weights, stack frames, and ring entries have polynomial bit length. Both recursions have logarithmic depth and retain only a polynomial number of such objects when streamed. This proves Theorem 1.2.

10 Comparison with compatible prior constructions

The five newest archives already combine much of the older work. Comparing only with the filed 76.909 coefficient would substantially overstate the new result. The strongest displayed cached certificate is the contraction-tree packet's $(35.354588, 34.274803)$ time–space pair; the strongest displayed polynomial-space certificate is the adaptive packet's 55.709478 [8, 7].

A stronger cached prior composition is also available. Use the paid singleton caches and recursive block-table construction from the tree packet, but choose a $3, 3, 2$ partition of eight groups minimizing crossing entropy as in the adaptive packet. Each directed edge is crossing with probability $3/4$, so the final direct crossing sum has coefficient

$$\beta_{\mathrm{old},8} = \tfrac{3}{4}\frac{\sqrt{56}}{8}.$$

Building the three blocks has smaller coefficient $\sqrt{766/25}/8$. At arities twelve and four retain the prior fixed three-block matrix construction, with coefficients $\sqrt{12/25}$ and $\sqrt{208/25}/4$, respectively. On our cached schedule the root dominates, with coefficient

$$C_{\mathrm{old,composed}} = \tfrac{3}{4}\frac{\sqrt{56}}{8}\,\kappa\sqrt{60} = 34.8321534202\ldots.$$

The root storage in this composition is $(\sqrt{30}/8)\kappa\sqrt{60} = 33.9927010637\ldots$. The supporting schedule checker recomputes these values rather than using the rounded text as a certificate. This composition is prior machinery, not counted as a new lemma.

Our cached bound is strictly below both the displayed and this composed time certificate, and uses a smaller space certificate. The analogous matched polynomial-space comparison was derived above. The guarantee is about the specified common interface. It does not numerically dominate unspecified constants in the separate Claude manuscript, low-degree bounds with stronger hypotheses, or unproved conditional order targets.

Running a finite portfolio of the new and inherited engines preserves the best valid bound in each input regime. For exact algorithms, ordinary round-robin simulation stops when the first finishes; its overhead is constant for a fixed portfolio. This also preserves special-case benefits whose assumptions differ from the general-degree normalization. We have not proved that our schedules are optimal over all possible portfolios or all future combinations of the underlying framework.

10.1 A slightly faster matrix backend

The parametric local theorem permits any established $7/3 \le \omega < 3$. Choosing $\omega = 19/8$ is supported by the primary bound in [11], with fixed slack. On the same schedule replace β by $\sqrt{617/1536}$. The

root table-formation estimates also decrease, so the same accounting applies. The exact expressions become

$$C_T(\omega) = \frac{\pi}{\ln 2}\frac{\sqrt{105}}{2} + \frac{\pi\sqrt{2}}{\ln 2}\frac{\sqrt{90}}{4}\sqrt{\frac{\omega^2+4}{24}}, \quad C_S(\omega) = \frac{\pi\sqrt{2}}{\ln 2}\sqrt{60}\sqrt{\frac{\omega^2+4}{24}}.$$

The main theorem uses 12/5 so that the principal comparison uses the same backend as the incoming tree packet. The optional refinement does not change the logarithmic order, and the code is not an implementation of either asymptotically fast backend. For $\omega = 19/8$, the numerical coefficients are $C_T = 32.8563452760\ldots$ and $C_S = 31.4674831881\ldots$.

11 Why the stronger logarithmic targets remain unresolved

11.1 Precision peeling is locally useful but not a new leading saving

The incoming frontier packet already implements computing a contribution of certified valuation V at residual precision $P - V$. It also explains that this does not reduce the leading recursive bound. Our audit confirms the reason: every active child creates a fresh Hermite loss of order MD/u, while inherited precision is only $O(\lambda^4 2^{-b_*})MD/u = o(MD/u)$. Reducing an already subordinate quantity leaves the leading term unchanged.

The peeling interface retains the same graph and ports. It does not correlate large mode weight with a smaller child wire count or lower deformation degree. Such an additional correlation could matter, but it has not been proved. Treating peeling alone as a second leading saving would repeat a limitation already documented in the incoming work.

11.2 A short index is not a bounded-degree tensor description

The precision packet compresses mode indices to $S_\beta = O(M/\sqrt{u} + r^2)$ rank bits. The compressed network still has only r genuine group tensors, of degree on the scale S_β/r. This fails the required sublinear maximum-degree condition when r is fixed or only polylogarithmic. Reapplying the same theorem is not licensed.

The quantitative tradeoff already in that packet is $u = O(\log M/\log r)$ for the first pass and $u' = O(\log r)$ for a second pass. Their putative savings multiply back to

$$\frac{M}{\sqrt{\log M/\log r}}\frac{1}{\sqrt{\log r}} = \frac{M}{\sqrt{\log M}}.$$

This is a barrier for that particular proof route, not a complexity lower bound for SAT or for every representation of these tensors.

11.3 The exterior-power route needs more than reduced degree

The earlier asymptotic-analysis packet explicitly asks for a closed recursive interface combining deformation degree $K(2L-1)$, $O(M/D)$ modes, $M/8$ compressed-wire children, and admissible descriptions and entry costs. If supplied, it would yield $\mathrm{poly}(s+\mathcal{L})2^{O(\sqrt{sK/\log s})}$, hence $2^{O(n\sqrt{\log\log n/\log n})}$ for its intended reduction.

The local exterior-power degree statement does not supply the required compressed mode count: the ordinary separation scales with the original incidence universe. Antisymmetry restricts occupation but

does not itself eliminate square-root partition entropy. A generic transfer of low incidence degree to rank-bit degree is also false: the singleton coefficient table of the degree-one polynomial z_{2^b} is AND on its b binary index bits, with Boolean degree b.

The sharp local Smith profiles in the newest kernel archive further limit improvements obtained just by increasing Cauchy-mode valuation weights. The present report gains by summing an existing exact mode graph more efficiently. It does not claim a superlinear valuation slope or a new compact tensor compiler.

12 Validation and reproduction

The archive includes the complete report source and compiled PDF; exact four-block and separator implementations; independent literal-sum tests over rings with zero divisors; rational schedule certificates; source and archive hashes; and a single replay command.

Check	What it establishes
Four-block certificate	All $6^3 = 216$ selections of cost forms, the analytic correction inequalities, and an exact equality witness.
Separator certificate	The sharp four-group inequality, hierarchical path bounds, exact ring contractions, and paid factor-call counts.
Cached graph evaluator	Equality with independent direct enumeration; all requested singleton entries counted; chosen pair contractions and cardinality-one index cases.
Schedule certificate	Exact capacity equalities, rational logarithmic gate, local build costs, interval-separated recurrence candidates, and eventual rounded coefficients.
PDF build	Compilation with shell escape disabled, resolved references, and visual inspection of rendered pages.

The executed checks include 216 exact cost-form selections, 46,656 exhaustive four-group weight vectors, 720 hierarchical path inequalities, 47 cached graph fixtures, and 216 streamed graph fixtures. The cached fixtures cover 17,174 complete assignments under independent literal enumeration; the streamed fixtures separately check 17,830 paid factor evaluations. The graph tests include modular rings and nonzero nilpotents. Exact interval arithmetic isolates every scalar schedule constant and proves the strict recurrence comparisons.

The recorded machine-readable results identify the deterministic seeds. Checks use explicit exceptions and remain active under `python -O`; normal and optimized results are compared. Finite fixtures validate the algebra and implementation interfaces. The proofs in the report establish the asymptotic claims. Neither kind of evidence is described as a new Lean/Rocq verification.

From the extracted archive, run

```
python3 code/replay.py
python3 code/build_report.py
```

The first command needs Python 3.10 or newer and its standard library. The second additionally needs pdfLaTeX and the packages listed in the preamble. The replay writes separate results instead of overwriting the delivered evidence. No network, credential, SAT repository build, or external service is

needed. The optional faster matrix backend is a theorem-level dependency, not a bundled production implementation.

A The twenty-packet review inventory

Packet	Relevant strongest contribution and status
`cauchy_mode_improvement`	Integral factorial modes with slope three; constant improvement, unchanged SAT order.
`eth_contraction_improvements`	Sharp Hermite condition counts, leading recovery precision six, streamed coefficient extraction.
`3sat_precision_refinement`	Exact recovery-loss and taper analysis; stronger $n \log\log n / \log n$ target conditional on missing summation.
`sat_density_improvement`	Uniform density-sensitive envelope (1); intermediate-density order improvement.
`eth_density_refinement`	Overlapping density refinement and precision/Smith analysis; no unrestricted order improvement.
`3sat_refinement`	Scalar parity lift and incidence-degree reduction; second-contraction target lacks the required compact factorization.
`3sat_asymptotic_analysis`	Scoped barriers and explicit missing degree-preserving and linear-wire interfaces.
`eth_asymmetric_packet_balancing`	Unequal retained/fresh degree bounds, general coefficient 76.908995; lower-degree results separate.
`3sat_round2_refinement`	Combined coefficient 84.06627; source and actual-domain audits.
`eth_balanced_packing`	Nearly lossless padding; coefficient 94.74820.
`eth_high_moment`	High-moment/orbit packing; same general coefficient 94.74820.
`eth_matching_refinement`	Matching-potential packing and thin precision gaps; same general coefficient 94.74820.
`eth_packet_refinement`	Pair-load control and thin precision gaps; coefficient 95.66394.
`eth_recursive_refinement`	Actual-domain modes and additive gaps; coefficient 133.20737.
`eth_joint_cauchy`	Full crossing-block slope-four modes; mode graph, cached triangle contraction; resource tradeoffs explicit.
`sat_adaptive_contraction`	Streamed separator coefficient 55.709478; cached adaptive coefficient 36.750313; low-degree and kernel benchmarks have different scopes.
`sat_frontier_improvements`	Variable-arity and cached compositions; practical kernel work. All retain the logarithmic order.
`subexponential_sat_frontier`	Multistage polynomial-space coefficient 63.852203; cached 37.930109; exact label scoring and precision peeling.
`subexponential_sat_kernel_improvements`	Exact local Smith profiles, adaptive elimination, packet extraction, and arithmetic improvements; no new global SAT order.
`subsat_contraction_tree`	Recursive cached table construction and coordinate/CRT matrix bridge; 35.354588 time, 34.274803 space; linear prefix feature generation.

The two primary manuscripts are the vendored OpenAI preprint and `Research/claude-2o-n-3sat/eth.pdf`. The former supplies the common normalized framework; the latter has the same stated general SAT order but no matched explicit coefficient for this common wire interface.

References

[1] Vladimir Reshetnikov, *ProveIt: SubexponentialSAT*, closing snapshot `dde01c4a63b8` (full identifier in the link). https://github.com/VladimirReshetnikov/ProveIt/tree/dde01c4a63b822c25b8eb345299346c36f837b49/Computability/SubexponentialSAT.

[2] *ProveIt incoming reports*, historical arrival snapshot `fa6e904d109b` (full identifier in the link). https://github.com/VladimirReshetnikov/ProveIt/tree/fa6e904d109b738154db7fe41e7f2777e6d24aee/docs/incoming.

[3] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. Upstream pin `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`; quantitative contraction and reduction appendices. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026.

[4] *3sat precision refinement*, first-round report, `Research/3sat_precision_refinement/report.tex` in [1].

[5] *Asymmetric packet balancing for subexponential SAT*, `Research/eth_asymmetric_packet_balancing/report.tex` in [1].

[6] *Joint Cauchy contraction*, `Research/eth_joint_cauchy/report.tex` in [1].

[7] *Adaptive aggregation for subexponential SAT*, `sat_adaptive_contraction_2026-10-26.zip`, internal `adaptive_sat_review.tex`, in [2]; now `Research/sat_adaptive_contraction/` in [1].

[8] *Contraction trees and streamed Cauchy features for subexponential SAT*, `subsat_contraction_tree_2026-10-26.zip`, internal `report/cached_tree.tex`, in [2]; now `Research/subsat_contraction_tree/` in [1].

[9] *Subexponential SAT: combined bounds and exact implementation improvements*, `subexponential_sat_frontier_2026-10-26.zip`, internal `theory.tex` and `implementation.tex`, in [2].

[10] *Subexponential SAT kernel improvements*, `subexponential_sat_kernel_improvements_2026-10-26.zip`, internal `report.tex`, in [2].

[11] Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349, SODA 2025. Only a fixed exponent below 19/8, and hence below 12/5, is used here; no current-record assertion is required. https://arxiv.org/abs/2404.16349.